Adding and subtracting time

1 Evaluate the following in hours and minutes:

- **a** 11 hours − 43 minutes
- **b** 11 hours 16 minutes − 5 hours
- **c** 5 hours 10 minutes + 20 minutes
- **d** 4 hours 10 minutes − 35 minutes
- **e** 3 hours 3 minutes + 3 hours 29 minutes
- **f** 3 hours 20 minutes − 2 hours 10 minutes
- **g** 2 hours 50 minutes + 3 hours 40 minutes
- **h** 6 hours 10 minutes − 2 hours 30 minutes
- **i** 7 hours 59 minutes + 12 hours 11 minutes
- **j** 10 hours 33 minutes − 7 hours 57 minutes
- **k** 10 hours 43 minutes − 2 hours 29 minutes

2 How many hours and minutes left until midnight if the time is 16:50?

3 Find the following times, expressing your answer in 24-hour time:

- **a** 4 hours after 16:28
- **b** 7 hours after 22:53
- **c** 12 hours before 9:57
- **d** 13 hours before 10:43
- **e** 30 minutes after 9:30 am.
- **f** 30 minutes after 11:40 pm.
- **g** 9:30 am plus half an hour.
- **h** 7:00 pm plus 24 minutes.
- **i** 45 minutes after 11:20 am.
- **j** 20 minutes after 3:50 pm.
- **k** 10:30 am plus two and a half hours.
- **l** 9:10 pm plus 63 minutes.

4 Find the 24-hour time and day after evaluating the following:

- **a** Saturday 13:15 − 15 hours 18 minutes
- **b** Thursday 13:20 + 21 hours 59 minutes

5 James went to a movie at 11:50 am. The movie went for 1 hour and 10 minutes. What time will be it when the movie finishes? Write your answer using 12-hour time.

6 Sharon starts watching a movie at half past eleven am, which goes for 2 hours and 30 minutes. What time will be it when the movie finishes? Write your answer using 12-hour time.

7 Valentina starts work at 7 am and finishes work at 5:30 in the evening. How many hours and minutes is Valentina at work?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

8 A train ride between two cities takes 9 hours and 57 minutes. On a train ride between these cities, if you leave one city at 23:44, at what time will you arrive at the other city? Give your answer in 24-hour time.

9 In a relay of four runners, Karen ran for 1:46, Janet ran for 1:18, Tania ran for 1:16 and Tina ran for 1:42. How long was their total time for the relay?

10 A ship arrives at 17:28 at Melbourne. If the trip took 2 hours and 16 minutes, at what time did the ship depart from Sydney? Express your answer in 24-hour time.

Time between events

11 James leaves home at 2 pm and gets home at 5 pm. How many hours was James out?

12 Luke starts school at 9 am and finishes at 3 pm. How many hours is Luke at school each day?

13 Ellie's family leaves her home at 9:17 am to visit her grandmother in a different state. She arrives at her grandmother's house at 3:17 pm. How long did the trip take?

14 A song clip starts playing at 7:55 pm and finished at 8:11 pm. How long is the song clip in minutes?

15 A taxi departs at 13:32 from Sydney and arrives at 20:14 at Melbourne. How long did the taxi take for the trip?

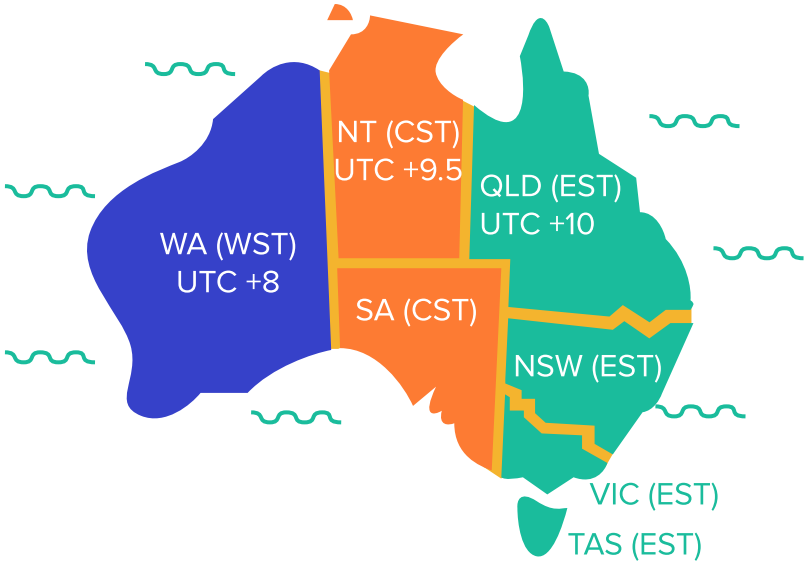
16 A match of hockey begins at 7:31 and finishes at 8:55. If the half-time break goes for 15 minutes, what is the duration of one half of hockey?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Time zones in Australia

17 Consider the map below:

Western Standard Time (WST) 2 hours behind EST	Central Standard Time (CST) 0.5 hours behind EST	Eastern Standard Time (EST)



a Determine the time in Perth when it is 6:42 pm in Hobart.

b Complete the table to show the equivalent time in each state:

Western Australia (WA)	South Australia (SA)	Queensland (QLD)
6:15		
		11:45
	10:15	
5:25		

c Complete the following table to show the equivalent time in each time zone:

WST	CST	EST
	5:25	
5:11		
		12:25
	7:17	

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

18 If it is 4:00 pm Daylight Savings Time (DST) in South Australia (SA), determine the time in:

a New South Wales (NSW).

b Western Australia (WA).

c Queensland.

d Northern Territory (NT).

19 Australia is divided into 3 time zones:

- AEST (Eastern Standard Time) encompasses QLD, NSW, the ACT, VIC and TAS.
- ACST (Central Standard Time) encompasses the NT and SA, and is half an hour behind AEST.
- AWST (Western Standard Time) encompasses WA and is two hours behind AEST.

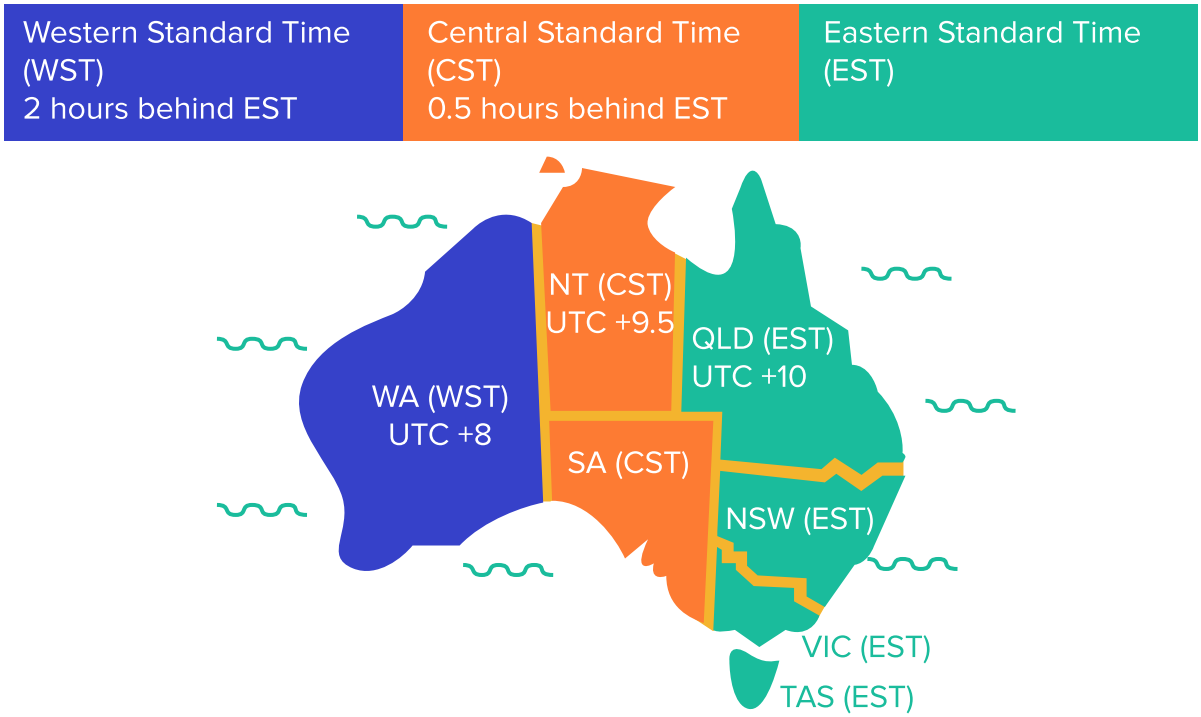
a If it is 12:10 am AEST, determine the following, giving your answers in 24-hour time:

i The time in ACST.

ii The time in AWST.

b If it is 1:16 pm in ACST, determine the time in AWST.

20 Han and Amelia board a train at 8:52 pm on Thursday from Sydney, New South Wales towards Adelaide, South Australia.



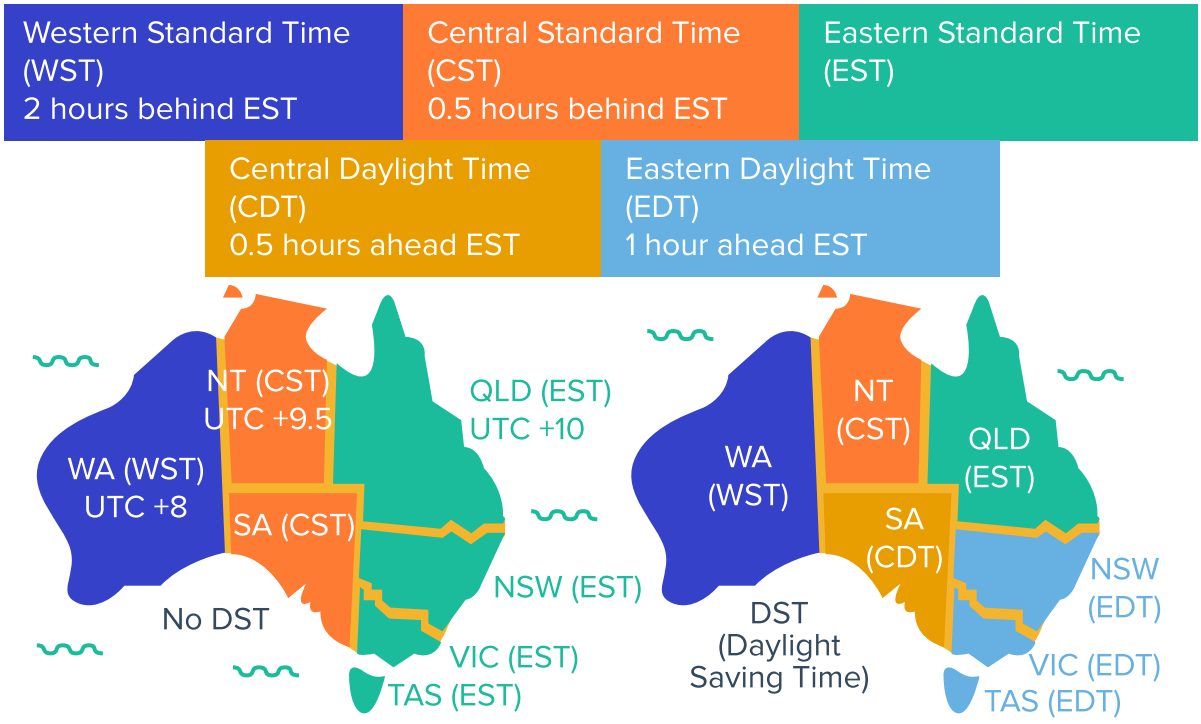
a They arrived in Adelaide, South Australia on 6:48 pm Friday. Determine the duration of their train trip.

b Han boards a train from Adelaide, South Australia at 8:43 pm on Friday to Darwin, Northern Territory. He arrives at 8:11 pm Saturday. Determine the duration of his trip.

c Amelia boards a train from Adelaide, South Australia at 7:53 pm Friday towards Perth, Western Australia. She arrives at 10:54 am. Determine the duration of her train trip.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

21 The map below shows the time zones across the Australian States in standard time and daylight saving time:



Calculate the flight times of the following trips:

a The airplane leaves Broome Airport in Western Australia at 7:30 pm and arrives at Albury (operating on daylight savings time) Airport in Victoria at 3:52 am.

b The airplane leaves Albury Airport in Victoria at 7:14 pm and arrives at Adelaide Airport in South Australia (operating on daylight savings time) at 9:10 pm.

c The airplane leaves Adelaide Airport in South Australia at 9:53 am and arrives at Broome Airport in Western Australia(operating in daylight savings time) at 3:41 pm.

22 If it is 16:14 in Sydney in Daylight Savings Time, determine the time it would be in Eastern Standard Time.

23 Determine the time it would be in Eastern Standard Time for the following:

a 2:49 am in Sydney in Daylight Savings Time.

b 11:50 pm in Sydney in Daylight Savings Time.

24 If it is 14:35 in Sydney in Eastern Standard Time, determine the time it would be in Daylight Savings Time.

25 Determine the time it would be in Daylight Saving Time for the following:

a 9:45 am in Sydney in Eastern Standard Time.

b 5:13 pm in Sydney in Eastern Standard Time.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

26 If it is 4 am in standard time, what time is it in daylight savings time (DST)?

27 If it is 8 am in daylight savings time (DST), what time is it in standard time?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647